

Document reference		NPS/002/003					
Version:-	1	Date of Issue:-	June 06	Page	1	of	8

NPS/002/003 – Specification for Cable Ducts, Protective Tiles and Tile Tape

1. Purpose

The purpose of this specification is to detail the technical requirements of CE Electric UK in relation to cable ducts used for buried cable installations, associated accessories, protective tiles and tile tape.

This document supersedes the following documents, all copies of which should be destroyed.

Ref	Version	Title
DSS/005/024	1	Specification For Service, LV and 11kV Mains Cable PVC Ducting
DSS/005/010	1	Specification for YE Plastic Marker Tape for Buried Electricity Cables
DSS/005/018		Specification for Service LV and HV Mains Cable PVC Ducting
MS-0/07		Specification for Cable Guards and Plastic Cable Covers

2. Scope

This document applies to the supply of PVC ducting and protective tile and tile tape for use on the CE Electric UK LV and HV underground distribution network.

3. Requirements

3.1 uPVC HV Cable Ducts, HDPE/MDPE HV Solid Wall Cable Ducts

The wall thickness of the cable duct shall be such as to meet the following tests and requirements:

- Class 1:
All ducts (uPVC & HDPE) intended for Low voltage mains distribution cables, for Medium voltage power cables and for High voltage power cables to give 450N compression strength at 75°C
- Class 2:
All ducts (uPVC & MDPE) intended for low voltage service cables and pilot/telephone cables to give 450N compression strength at 50°C

As the design stresses and the physical properties of uPVC and HDPE are different, duct bore/wall thickness ratios will be different.

All ducts that are to be buried shall retain their shape (less than 5% deformation) under crushable pressure of 750N/m for class 1 and in excess of 450N/m for class 2 as detailed in ST4 of ENATS 12-24.

Document reference		NPS/002/003					
Version:-	1	Date of Issue:-	June 06	Page	2	of	8

All ducts shall be indelibly and clearly marked with the legend 'ELECTRIC CABLE DUCT' in accordance with the ENATS 12-24.

3.2 Ducts

All ducts shall be as detailed in ENATS 12-24 and to this specification, duct requirements are: -

- a. Ducts required for High voltage power cables (class 1) shall be non-coilable/semi-rigid in either "twin walled" or with a smooth outer wall, manufactured in either uPVC or HDPE; each duct type shall have a smooth inner surface.
- b. Duct required for low voltage service cables and pilot/telephone cables (class 2) shall be uPVC or MDPE, smoothed walled with a smooth inner surface.

Bore and duct length are as detailed in Appendix 1.

The cross section of all ducts shall be circular with the ends of the ducts cleanly cut and square with the longitudinal axis of the duct, they shall have no sharp edges, burrs or surface projections which are likely to damage the insulation of cables and shall not present any impedance to the installation or withdrawal of cable throughout its length.

Each duct length shall be supplied with a straight coupler incorporated in one end and designed to be a close push fit onto a plain end of a duct with the same nominal diameter. The duct ends shall be suitable for application without the use of adhesive materials.

3.3 "Slow" Radius Bends

HDPE, MDPE and uPVC bends shall have either a smooth outer and inner surface, twin walled bend shall have a corrugated outer and a smooth inner surface. Each bend shall incorporate a coupler in one end and shall be suitable for application without the use of adhesive materials. The bend shall not present any internal projection or impedance to the installation or withdrawal of cables.

3.4 Straight Couplings

Straight couplings shall comply with requirements of ENATS 12-24 and shall be similar to the 'Typical Couplers' shown in section 5 of that standard. All duct couplings shall not present any internal projection or impedance to the installation or withdrawal of cable through the duct run couplings shall be suitable for application without the use of adhesive materials.

3.5 Hockey Stick

The 'Hockey Stick' shall be manufactured from uPVC and have an internal diameter of 38mm and to the requirements of ENATS 12-24. The end of the hockey stick shall be so designed to be a close push fit into the end of a duct or bend with the same nominal diameter.

After installation, the "long" end of the hockey stick can be expected to be subjected to the full range of climatic conditions encountered in the UK and may be exposed to sunlight for a significant period.

3.6 Adaptors

Adaptors shall be supplied for straight coupling of the 38mm internal diameter ducts to the 50mm internal diameter ducts Adaptors shall be suitable for application without the use of adhesive materials.

3.7 Cable cover for LV and HV Cables

The tile will be made in such a way to give a clear visual warning to contractors and Utilities of the presence of underground cables or ducts during excavation.

Document reference		NPS/002/003					
Version:-	1	Date of Issue:-	June 06	Page	3	of	8

The tile shall be manufactured from reconstituted low or medium density Polythene or a similar material, which shall be rot-proof and resistant to a wide variety of virgin and tipped soil conditions. The ends of the tile shall be cleanly cut, square with its longitudinal axis and shall have no sharp edges. The tile shall be laminated with suitable marker tape identification complying with ENATS standard 12-23 and meet impact requirements of BS-2484.

Nominal length shall be 40M +/- 1M.

3.8 Cable Tiles for use on 33kV and 66kV Cables

The tile shall be made in such a way to give a clear visual warning to contractors and Utilities of the presence of underground cables or ducts during excavation.

The tile shall meet impact requirements of BS-2484, comply with ENATS 12-23 and manufactured from any thermoplastic material providing it conforms to the requirements of the relevant standard. The use of reworked, reprocessed or recycled materials is permitted. However, the material used shall have no detrimental effects on the environment and shall be rot-proof and resistant to a wide variety of virgin and tipped soil conditions.

The tile shall have a suitable standard warning text laminated along its longitudinal length complying with ENATS 12-23.

The tile ends shall be cleanly cut, square with the longitudinal axis of the tile and shall have no sharp edges, shall be drilled with a 12mm hole with a centre 150mm from the side and 25mm from the end with a suitable peg or cable tie provided to enable tiles to be connected together.

Document reference		NPS/002/003					
Version:-	1	Date of Issue:-	June 06	Page	4	of	8

4. References

4.1 External Documentation

Reference	Title
ENATS 12 – 23 Issue 2 - 2004	Polyethylene Protection Tape for Buried Electricity Supply Cable
ENATS 12 – 24 Issue 2 - 2005 (draft)	Plastic Ducts for Buried Electric Cables
BS EN 50086	Conduit Systems for Cable Management (Part 1-3)
BS EN 12613	Plastic Warning Devices for Underground Cables and Pipelines with Visual Characteristics
BS 2484	Specification for straight concrete and clay-ware cable covers

4.2 Internal Documentation

Reference	Title
Nil	

5. Definitions

Term	Definition
Nil	

Document reference		NPS/002/003					
Version:-	1	Date of Issue:-	June 06	Page	5	of	8

6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
R McMahon	Standards Engineer	R McMahon	23/05/06

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
G Bolden	Procurement Manager	G Bolden	23/05/06

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
D M Gazda	Underground Standards and Assessment Manager	D M Gazda	23/05/06

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S Johnson	DBD Administrator	S Johnson	02/06/06

6.5 Approval

Approval is given for the content of this document

		Sign	Date
C Holdsworth	Standards and Assessments Manager	C Holdsworth	23/05/06

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
C Morgan	Head of Procurement	C Morgan	25/05/06

Document reference		NPS/002/003					
Version:-	1	Date of Issue:-	June 06	Page	6	of	8

Appendix 1 – Current Requirements

Item	Description	Colour of Product	Commodity Code
1	38mm(ID) x 5000mm uPVC Rigid Duct	Black	140348
2	38mm (ID) uPVC “Hockey Stick”	White	140225
3	38mm (ID) uPVC “Hockey Stick”	Black	TBA
4	38mm (ID) x 90° uPVC Bend	Black	140352
5	50mm(ID) x 1500mm uPVC Rigid Duct	Black	160888
6	50mm (ID) 90° uPVC Bend	Black	160835
7	50mm – 38mm uPVC Reducer / Coupler	Black	140333
8	100mm(min)(ID) x 3000mm uPVC Twin Walled	Black	262717
9	150mm(min)(ID) x 2000mm uPVC Twin Walled	Black	262535
10	150mm(min)(ID) x 6000mm uPVC Twin Walled	Black	262550
11	125mm(min) x 3000mm Twin Walled	Black	140314
12	Coupler for above	Black	140367
13	Twin Wall 125mm(min) x 90° Slow Bend	Black	140371
14	Twin Wall SPLIT 125mm(min) x 100mm	Black	140386
15	Cable Tile Tape: 200mm x 40M x 2.5mm	--	140761
16	Cable Tile Tape: 150mm x 40M x 2.5mm	--	140765
17	Cable Tile c/w Pegs : 1000mm x 244mm x 12mm	--	261925

Document reference		NPS/002/003					
Version:-	1	Date of Issue:-	June 06	Page	7	of	8

Appendix 2 – Addendum to Supplier Requirements

To enable CE Electric UK to install the product(s) in accordance with the manufacturer's recommendations, if appropriate, the supplier shall provide a single copy of drawings, descriptive leaflets and instruction manuals appropriate to the goods being offered, which shall incorporate details of the recommended installation procedures required in order to provide optimal performance of the goods during their operational life.

Document reference		NPS/002/003					
Version:-	1	Date of Issue:-	June 06	Page	8	of	8

Appendix 3 – Logistical Requirements

Palletised goods shall be supplied on standard 1200mm x 1000mm pallets.

Clearly legible, easily identifiable, durable and unambiguous labelling shall be applied to each individual and where relevant multiple packaged products. As a minimum requirement the following shall be included: -

- ❖ Manufacturer's trade mark or name
- ❖ Supplier's trademark or name
- ❖ Description of item
- ❖ Date of packaging and/or batch number
- ❖ CE-Electric UK brand or name
- ❖ CE-Electric UK product code
- ❖ Weight